

1. Introduction:

To analyze customer transactions and reviews to derive meaningful business insights, integrate structured and unstructured datasets, and develop an Ontology to model relationships between customers and products.

Datasets:

Structured: Customer transactions in CSV format (transaction amounts, product IDs etc).

Unstructured: Customer reviews in JSON format (review text, ratings etc).

2. Methodology:

This section outlines the process followed to integrate datasets, create an RDF ontology, validate it, and visualize the relationships as a graph.

Installed necessary library like rdflib, pandas etc and Uploaded the structure and unstructured data in google Colab.

Data Integration: A structured dataset containing Customer ID, Transaction Amount, Product ID, and Transaction Date. An unstructured dataset containing customer reviews and ratings.

To combine insights from structured and unstructured datasets to derive a holistic understanding of customer behavior and preferences.

Data Analysis: The structured dataset was analyzed to: Average transaction amount. The unstructured dataset was analyzed to: Identify the distribution of customer ratings.

After entered code in google Colab output saved into pdf (File > Download > PDF) and output.xml file downloaded in that RDF/XML format.

Validation and Visualization:

RDF Validation: The RDF/XML file validate successfully by using the W3C RDF Validator. After serialized the RDF format and validated that rdf structure is Validated.

Graph Visualization: A network graph was generated to visualize relationships between customers and products.

Ontology Development:

Entities: Customer, Product.

Relationships: buys: Links a customer to the product they purchased.

A customer Ontology was created, representing customers and their purchases as triples in the format:

<Customer> buys <Product>.

Diagram (1) represented the <Customer> buys <Product> relationship. In the diagram buys as represented by owl: things and customer as represented by persons as shown in the Diagram (1).

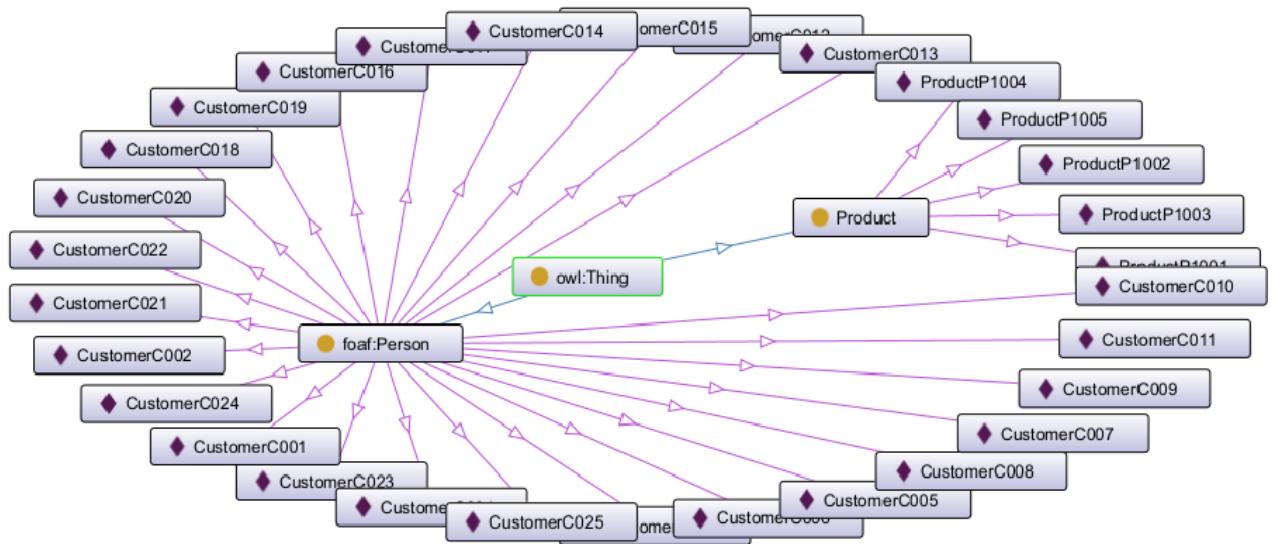


Diagram (1)

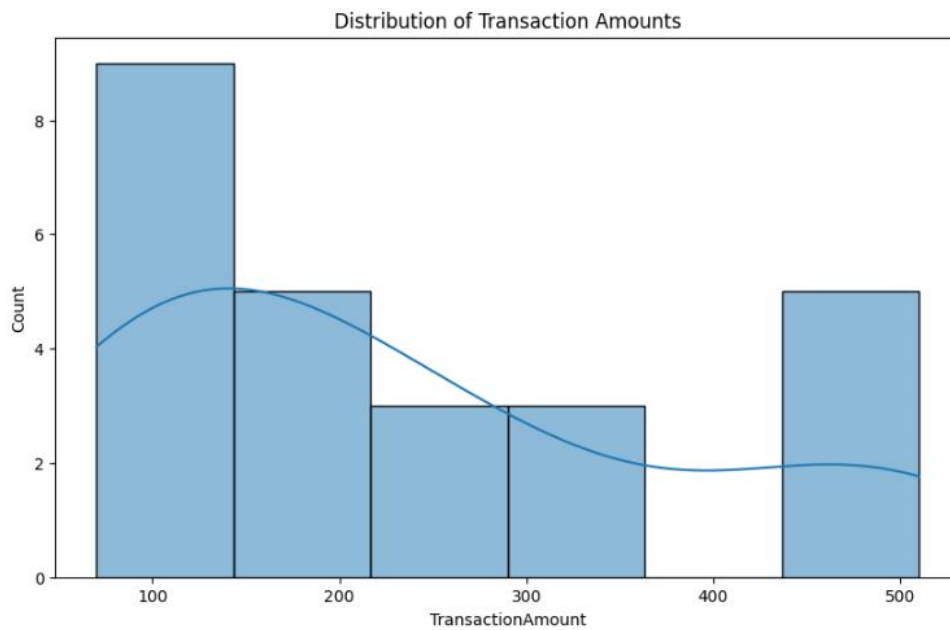
3. Results:

Average Transaction Amount: 231.58879999999996

Most Common Customer Rating: 5

Graphs:

1. Distribution of Transaction Amount:



Diagram(2)

2. Distribution of Customer Rating:

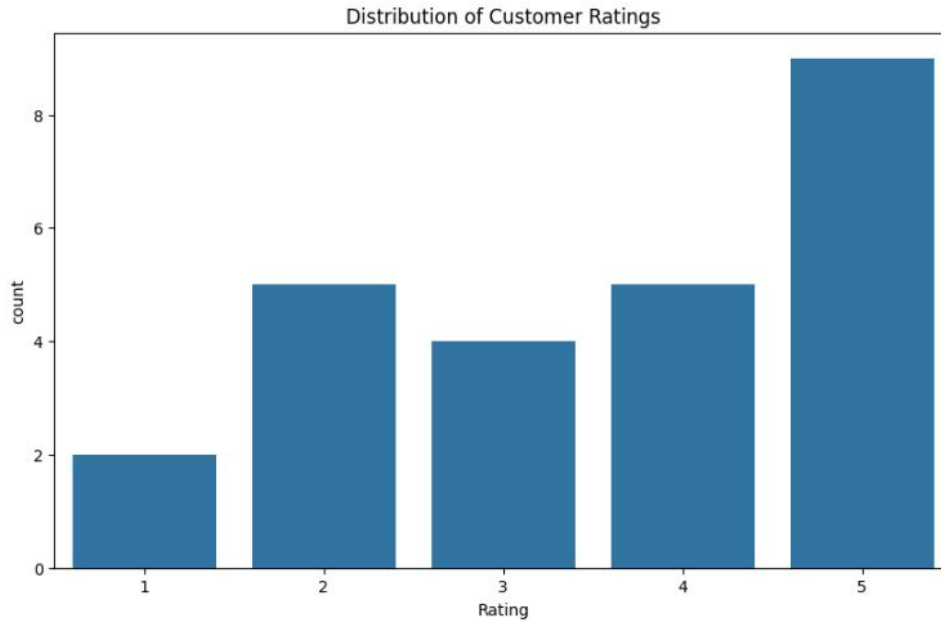


Diagram (3)

1. Google Colab notebook attached.
2. Output.xml (RDF/XML file) attached.

RDF Validation:

The serialized RDF was uploaded to the W3C RDF Validator, a tool used to check for errors and validate the syntax.

1. W3C RDF Validator document is attached.

Serialized rdf structure :

@prefix ns1: <http://example.org/> .

@prefix foaf: <http://xmlns.com/foaf/0.1/> .

ns1:CustomerC003 a foaf:Person ;

ns1:buys ns1:ProductP1002 .

ns1:CustomerC018 a foaf:Person ;

ns1:buys ns1:ProductP1003 .

ns1:CustomerC010 a foaf:Person ;

ns1:buys ns1:ProductP1005 .

ns1:CustomerC008 a foaf:Person ;
 ns1:buys ns1:ProductP1003 .

ns1:CustomerC013 a foaf:Person ;
 ns1:buys ns1:ProductP1002 .

ns1:CustomerC021 a foaf:Person ;
 ns1:buys ns1:ProductP1001 .

ns1:CustomerC024 a foaf:Person ;
 ns1:buys ns1:ProductP1004 .

ns1:CustomerC004 a foaf:Person ;
 ns1:buys ns1:ProductP1004 .

ns1:ProductP1001 a ns1:Product .

ns1:CustomerC016 a foaf:Person ;
 ns1:buys ns1:ProductP1001 .

ns1:CustomerC015 a foaf:Person ;
 ns1:buys ns1:ProductP1005 .

ns1:ProductP1005 a ns1:Product .

ns1:CustomerC006 a foaf:Person ;
 ns1:buys ns1:ProductP1001 .

ns1:ProductP1004 a ns1:Product .

ns1:CustomerC017 a foaf:Person ;
 ns1:buys ns1:ProductP1002 .

ns1:CustomerC022 a foaf:Person ;
 ns1:buys ns1:ProductP1003 .

ns1:CustomerC023 a foaf:Person ;
 ns1:buys ns1:ProductP1002 .

ns1:CustomerC005 a foaf:Person ;
 ns1:buys ns1:ProductP1005 .

ns1:CustomerC001 a foaf:Person ;
 ns1:buys ns1:ProductP1001 .

ns1:CustomerC012 a foaf:Person ;
 ns1:buys ns1:ProductP1003 .

ns1:CustomerC011 a foaf:Person ;
 ns1:buys ns1:ProductP1001 .

ns1:CustomerC019 a foaf:Person ;
 ns1:buys ns1:ProductP1004 .

ns1:CustomerC020 a foaf:Person ;
 ns1:buys ns1:ProductP1005 .

ns1:CustomerC007 a foaf:Person ;
 ns1:buys ns1:ProductP1002 .

ns1:CustomerC025 a foaf:Person ;
 ns1:buys ns1:ProductP1005 .

ns1:ProductP1002 a ns1:Product .

ns1:ProductP1003 a ns1:Product .

ns1:CustomerC014 a foaf:Person ;
 ns1:buys ns1:ProductP1004 .

ns1:CustomerC002 a foaf:Person ;
 ns1:buys ns1:ProductP1003 .

ns1:CustomerC009 a foaf:Person ;
 ns1:buys ns1:ProductP1004 .

Ontology: The Ontology illustrates relationships between customers and products using triples.

Customers are modeled as **foaf:Person**, and products are instances of **ns1:Product**. The **ns1: buys** predicate connects customers to the products they purchase, forming a graph that highlights individual transactions and overall purchase patterns.

Customers (foaf:Person): Represent individual buyers, CustomerC001 to CustomerC0025.

Products (ns1:Product): Represent the items purchased, ProductP1001 to ProductP1005.

Edges Only Connect Two Sets: Customers are only connected to products via the ns1:buys predicate.

No Customer-to-Customer Connections: Customers are independent and have no direct relationships with each other.

No Product-to-Product Connections: Products are independent and not related unless indirectly through shared customers. As below shown in the Diagram (4) Ontology.

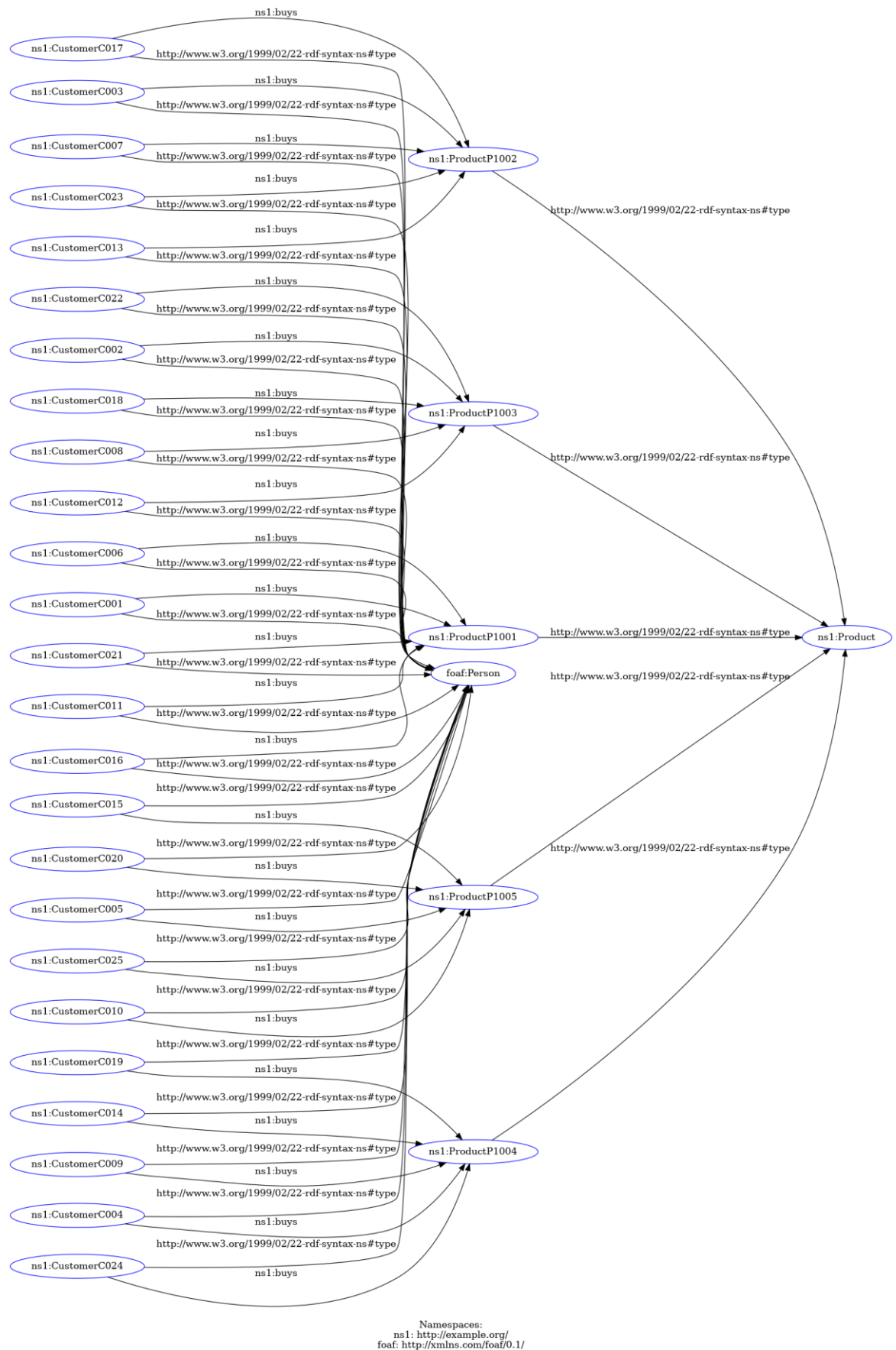


Diagram (4)

4. Discussion:

Findings:

Analyzing the graph reveals Purchased Products:

ProductP1001: Purchased by 5 customers (CustomerC001, CustomerC006, CustomerC011, CustomerC016, CustomerC021).

ProductP1002: Purchased by 5 customers (CustomerC003, CustomerC007, CustomerC013, CustomerC017, CustomerC023).

ProductP1003: Purchased by 5 customers (CustomerC002, CustomerC008, CustomerC012, CustomerC018, CustomerC022).

ProductP1004: Purchased by 5 customers (CustomerC004, CustomerC009, CustomerC014, CustomerC019, CustomerC024).

ProductP1005: Purchased by 5 customers (CustomerC005, CustomerC010, CustomerC015, CustomerC020, CustomerC025).

Each product (ProductP1001, ProductP1002, ProductP1003, ProductP1004, and ProductP1005) is purchased by exactly five unique customers.

No product is purchased disproportionately more than others.

This reflects an equal distribution of purchases across all products in the dataset, so there are no discernible "most popular" products.

These clusters highlight groupings of customers tied to specific products, but with no overlap between clusters (no customer is linked to more than one product).

One-to-One Relationships:

Each customer is linked to exactly one product, and there are no repeated or overlapping customer-product relationships.

This behavior contrasts with real-world scenarios where customers often buy multiple products or make repeat purchases.

Implications:

Business Insights: Equal product distribution reflects fairness but doesn't represent real-world trends like popular or high-demand items. Single-purchase behavior limits understanding of complex customer habits.

Ontology Benefits: Effectively integrates diverse data types (structured and unstructured) and scalable for more complex datasets and relationships.

Data Visualization: Ontology provide clear insights into customer-product links but lack complexity due to no overlapping relationships.

Future Recommendations:

Extend Attributes for customers (e.g., age, location) and products (e.g., category, price).

Introduce time-based data to analyze purchase trends, seasonality, and repeat customer behavior.

Use customer reviews (sentiment analysis) and structured transaction data for clustering or predictive modeling of customer preferences.

5. Conclusion:

This analysis demonstrated the use of Ontology to integrate structured transaction data and unstructured reviews, effectively modeling and visualizing customer-product relationships.

The RDF graph provided clear one-to-one links but lacked real-world complexity like repeat purchases or multi-product behaviors.

Despite the dataset's simplicity, the framework shows potential for tasks such as customer segmentation and recommendation systems.

Future work should incorporate more dynamic and realistic data for deeper insights and broader applications.

6. References:

W3C RDF Validator: RDF Schema Standards

Protégé: ontology

```
%pip install rdflib
```

```
Collecting rdflib
  Downloading rdflib-7.1.1-py3-none-any.whl.metadata (11 kB)
Collecting isodate<1.0.0,>=0.7.2 (from rdflib)
  Downloading isodate-0.7.2-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: pyparsing<4,>=2.1.0 in /usr/local/lib/python3.10/dist-packages (from rdflib) (3.2.0)
Downloading rdflib-7.1.1-py3-none-any.whl (562 kB)
562.4/562.4 kB 8.7 MB/s eta 0:00:00
Downloading isodate-0.7.2-py3-none-any.whl (22 kB)
Installing collected packages: isodate, rdflib
Successfully installed isodate-0.7.2 rdflib-7.1.1
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from rdflib import Graph, Literal, RDF, URIRef
from rdflib.namespace import FOAF, XSD
```

```
# Sample data paths (replace with actual paths)
csv_file_path = 'customer_transactions.csv'
json_file_path = 'customer_reviews.json'
```

```
# Load structured data (CSV)
structured_data = pd.read_csv(csv_file_path)
# Load unstructured data (JSON)
unstructured_data = pd.read_json(json_file_path)
```

```
# Explore structured data
print(structured_data.head())
print(structured_data.describe())
# Explore unstructured data
print(unstructured_data.head())
```

```
CustomerID  TransactionAmount  ProductID  Date
0          C001             150.00     P1001  1/5/2023
1          C002             200.50     P1003  1/6/2023
2          C003              99.99     P1002  1/7/2023
3          C004             300.00     P1004  1/8/2023
4          C005             450.75     P1005  1/9/2023

TransactionAmount
count      25.000000
mean       231.588800
std        148.590378
min         70.000000
25%         99.990000
50%        200.500000
75%        320.000000
max        510.000000

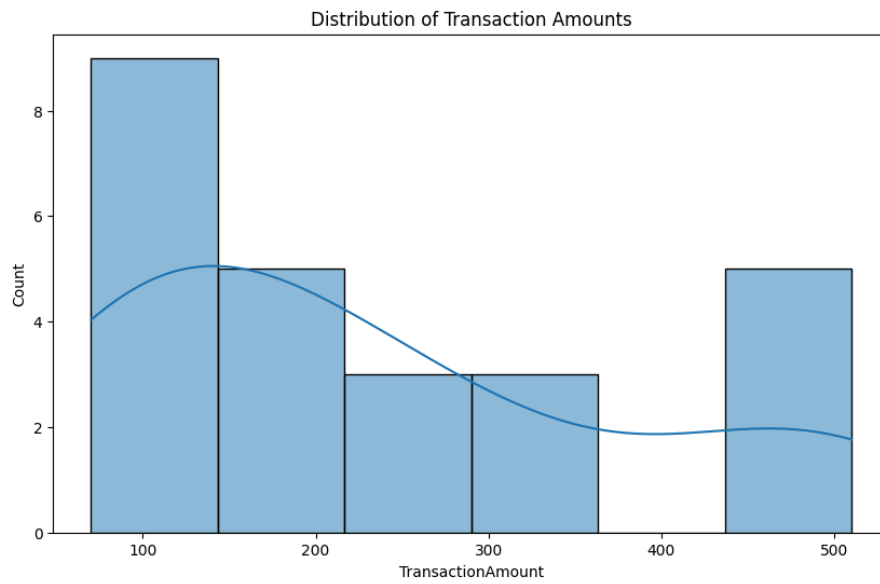
CustomerID  ReviewText  Rating
0          C001  Great product quality!  5
1          C002  Satisfied with the purchase.  4
2          C003  Could be better.  3
3          C004  Excellent service and product.  5
4          C005  Not what I expected.  2
```

```
# Handling missing values in structured data
structured_data.fillna(method='ffill', inplace=True)
```

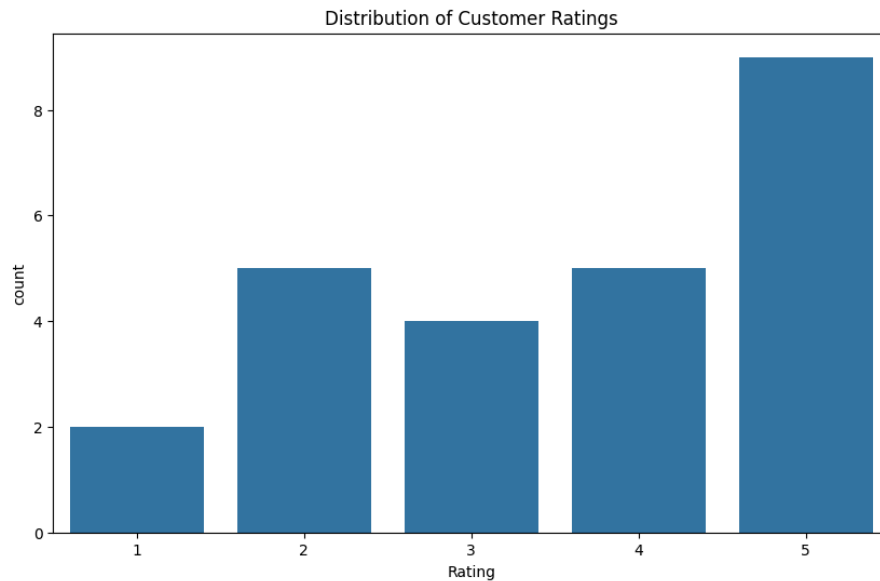
```
<ipython-input-6-47df102b0284>:2: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use ob
structured_data.fillna(method='ffill', inplace=True)
```

```
# Basic text preprocessing for unstructured data
unstructured_data['ReviewText'] = unstructured_data['ReviewText'].str.lower()
```

```
# Visualizing distribution of transaction amounts
plt.figure(figsize=(10, 6))
sns.histplot(structured_data['TransactionAmount'], kde=True)
plt.title('Distribution of Transaction Amounts')
plt.show()
```



```
# Visualizing ratings from reviews
plt.figure(figsize=(10, 6))
sns.countplot(x='Rating', data=unstructured_data)
plt.title('Distribution of Customer Ratings')
plt.show()
```



```
# Create a basic ontology graph
g = Graph()
```

```

# Define a namespace
namespace = URIRef("http://example.org/")

# Add triples to the graph
for _, row in structured_data.iterrows():
    customer = URIRef(namespace + "Customer" + str(row['CustomerID']))
    product = URIRef(namespace + "Product" + str(row['ProductID']))
    g.add((customer, RDF.type, FOAF.Person))
    g.add((product, RDF.type, URIRef(namespace + "Product")))
    g.add((customer, URIRef(namespace + "buys"), product))

print(g.serialize(format='xml'))

<?xml version="1.0" encoding="utf-8"?>
<rdf:RDF
  xmlns:ns1="http://example.org/"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
>
  <rdf:Description rdf:about="http://example.org/CustomerC008">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC015">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC018">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC001">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1001"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/ProductP1005">
    <rdf:type rdf:resource="http://example.org/Product"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC007">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC010">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC017">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC022">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC004">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC012">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC019">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC002">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC023">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
  </rdf:Description>

# Example: Average transaction amount
avg_transaction = structured_data['TransactionAmount'].mean()
print(f"Average Transaction Amount: {avg_transaction}")

Average Transaction Amount: 231.58879999999996

```

```
# Example: Most common rating
most_common_rating = unstructured_data['Rating'].mode()[0]
print(f"Most Common Rating: {most_common_rating}")
```

↪ Most Common Rating: 5

```
with open('output.xml', 'w') as f:
    f.write(g.serialize(format='xml'))
```

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This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<rdf:RDF xmlns:ns1="http://example.org/" xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
  <rdf:Description rdf:about="http://example.org/CustomerC021">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1001"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC007">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/ProductP1004">
    <rdf:type rdf:resource="http://example.org/Product"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC020">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC003">
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  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/ProductP1005">
    <rdf:type rdf:resource="http://example.org/Product"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC018">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC025">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/ProductP1003">
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  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC004">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC015">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC012">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC023">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC009">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC024">
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    <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
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  <rdf:Description rdf:about="http://example.org/CustomerC017">
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  </rdf:Description>
  <rdf:Description rdf:about="http://example.org/CustomerC022">
    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
  </rdf:Description>
</rdf:RDF>
```

[illegible]

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Validation Results

Your RDF document validated successfully.

Triples of the Data Model

Number	Subject	Predicate	Object
1	http://example.org/CustomerC003	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
2	http://example.org/CustomerC003	http://example.org/buys	http://example.org/ProductP1002
3	http://example.org/CustomerC018	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
4	http://example.org/CustomerC018	http://example.org/buys	http://example.org/ProductP1003
5	http://example.org/CustomerC010	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
6	http://example.org/CustomerC010	http://example.org/buys	http://example.org/ProductP1005
7	http://example.org/CustomerC008	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
8	http://example.org/CustomerC008	http://example.org/buys	http://example.org/ProductP1003
9	http://example.org/CustomerC013	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
10	http://example.org/CustomerC013	http://example.org/buys	http://example.org/ProductP1002
11	http://example.org/CustomerC021	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
12	http://example.org/CustomerC021	http://example.org/buys	http://example.org/ProductP1001
13	http://example.org/CustomerC024	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
14	http://example.org/CustomerC024	http://example.org/buys	http://example.org/ProductP1004
15	http://example.org/CustomerC004	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
16	http://example.org/CustomerC004	http://example.org/buys	http://example.org/ProductP1004
17	http://example.org/ProductP1001	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://example.org/Product
18	http://example.org/CustomerC016	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
19	http://example.org/CustomerC016	http://example.org/buys	http://example.org/ProductP1001
20	http://example.org/CustomerC015	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
21	http://example.org/CustomerC015	http://example.org/buys	http://example.org/ProductP1005
22	http://example.org/ProductP1005	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://example.org/Product
23	http://example.org/CustomerC006	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
24	http://example.org/CustomerC006	http://example.org/buys	http://example.org/ProductP1001
25	http://example.org/ProductP1004	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://example.org/Product
26	http://example.org/CustomerC017	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
27	http://example.org/CustomerC017	http://example.org/buys	http://example.org/ProductP1002
28	http://example.org/CustomerC022	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
29	http://example.org/CustomerC022	http://example.org/buys	http://example.org/ProductP1003

30	http://example.org/CustomerC023	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
31	http://example.org/CustomerC023	http://example.org/buys	http://example.org/ProductP1002
32	http://example.org/CustomerC005	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
33	http://example.org/CustomerC005	http://example.org/buys	http://example.org/ProductP1005
34	http://example.org/CustomerC001	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
35	http://example.org/CustomerC001	http://example.org/buys	http://example.org/ProductP1001
36	http://example.org/CustomerC012	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
37	http://example.org/CustomerC012	http://example.org/buys	http://example.org/ProductP1003
38	http://example.org/CustomerC011	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
39	http://example.org/CustomerC011	http://example.org/buys	http://example.org/ProductP1001
40	http://example.org/CustomerC019	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
41	http://example.org/CustomerC019	http://example.org/buys	http://example.org/ProductP1004
42	http://example.org/CustomerC020	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
43	http://example.org/CustomerC020	http://example.org/buys	http://example.org/ProductP1005
44	http://example.org/CustomerC007	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
45	http://example.org/CustomerC007	http://example.org/buys	http://example.org/ProductP1002
46	http://example.org/CustomerC025	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
47	http://example.org/CustomerC025	http://example.org/buys	http://example.org/ProductP1005
48	http://example.org/ProductP1002	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://example.org/Product
49	http://example.org/ProductP1003	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://example.org/Product
50	http://example.org/CustomerC014	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
51	http://example.org/CustomerC014	http://example.org/buys	http://example.org/ProductP1004
52	http://example.org/CustomerC002	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
53	http://example.org/CustomerC002	http://example.org/buys	http://example.org/ProductP1003
54	http://example.org/CustomerC009	http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://xmlns.com/foaf/0.1/Person
55	http://example.org/CustomerC009	http://example.org/buys	http://example.org/ProductP1004

The original RDF/XML document

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1: <rdf:RDF xmlns:ns1="http://example.org/" xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
2:   <rdf:Description rdf:about="http://example.org/CustomerC003">
3:     <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
4:     <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
5:   </rdf:Description>
6:   <rdf:Description rdf:about="http://example.org/CustomerC018">
7:     <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
8:     <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
9:   </rdf:Description>
10:  <rdf:Description rdf:about="http://example.org/CustomerC010">
11:    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
12:    <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
13:  </rdf:Description>
14:  <rdf:Description rdf:about="http://example.org/CustomerC008">
15:    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
16:    <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
17:  </rdf:Description>
18:  <rdf:Description rdf:about="http://example.org/CustomerC013">
19:    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
20:    <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
21:  </rdf:Description>
22:  <rdf:Description rdf:about="http://example.org/CustomerC021">
23:    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
24:    <ns1:buys rdf:resource="http://example.org/ProductP1001"/>
25:  </rdf:Description>
26:  <rdf:Description rdf:about="http://example.org/CustomerC024">
27:    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
28:    <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
29:  </rdf:Description>
30:  <rdf:Description rdf:about="http://example.org/CustomerC004">
31:    <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
32:    <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
33:  </rdf:Description>
34:  <rdf:Description rdf:about="http://example.org/ProductP1001">
35:    <rdf:type rdf:resource="http://example.org/Product"/>

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36: </rdf:Description>
37: <rdf:Description rdf:about="http://example.org/CustomerC016">
38: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
39: <ns1:buys rdf:resource="http://example.org/ProductP1001"/>
40: </rdf:Description>
41: <rdf:Description rdf:about="http://example.org/CustomerC015">
42: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
43: <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
44: </rdf:Description>
45: <rdf:Description rdf:about="http://example.org/ProductP1005">
46: <rdf:type rdf:resource="http://example.org/Product"/>
47: </rdf:Description>
48: <rdf:Description rdf:about="http://example.org/CustomerC006">
49: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
50: <ns1:buys rdf:resource="http://example.org/ProductP1001"/>
51: </rdf:Description>
52: <rdf:Description rdf:about="http://example.org/ProductP1004">
53: <rdf:type rdf:resource="http://example.org/Product"/>
54: </rdf:Description>
55: <rdf:Description rdf:about="http://example.org/CustomerC017">
56: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
57: <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
58: </rdf:Description>
59: <rdf:Description rdf:about="http://example.org/CustomerC022">
60: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
61: <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
62: </rdf:Description>
63: <rdf:Description rdf:about="http://example.org/CustomerC023">
64: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
65: <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
66: </rdf:Description>
67: <rdf:Description rdf:about="http://example.org/CustomerC005">
68: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
69: <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
70: </rdf:Description>
71: <rdf:Description rdf:about="http://example.org/CustomerC001">
72: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
73: <ns1:buys rdf:resource="http://example.org/ProductP1001"/>
74: </rdf:Description>
75: <rdf:Description rdf:about="http://example.org/CustomerC012">
76: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
77: <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
78: </rdf:Description>
79: <rdf:Description rdf:about="http://example.org/CustomerC011">
80: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
81: <ns1:buys rdf:resource="http://example.org/ProductP1001"/>
82: </rdf:Description>
83: <rdf:Description rdf:about="http://example.org/CustomerC019">
84: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
85: <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
86: </rdf:Description>
87: <rdf:Description rdf:about="http://example.org/CustomerC020">
88: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
89: <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
90: </rdf:Description>
91: <rdf:Description rdf:about="http://example.org/CustomerC007">
92: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
93: <ns1:buys rdf:resource="http://example.org/ProductP1002"/>
94: </rdf:Description>
95: <rdf:Description rdf:about="http://example.org/CustomerC025">
96: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
97: <ns1:buys rdf:resource="http://example.org/ProductP1005"/>
98: </rdf:Description>
99: <rdf:Description rdf:about="http://example.org/ProductP1002">
100: <rdf:type rdf:resource="http://example.org/Product"/>
101: </rdf:Description>
102: <rdf:Description rdf:about="http://example.org/ProductP1003">
103: <rdf:type rdf:resource="http://example.org/Product"/>
104: </rdf:Description>
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106: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
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108: </rdf:Description>
109: <rdf:Description rdf:about="http://example.org/CustomerC002">
110: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
111: <ns1:buys rdf:resource="http://example.org/ProductP1003"/>
112: </rdf:Description>
113: <rdf:Description rdf:about="http://example.org/CustomerC009">
114: <rdf:type rdf:resource="http://xmlns.com/foaf/0.1/Person"/>
115: <ns1:buys rdf:resource="http://example.org/ProductP1004"/>
116: </rdf:Description>
117: </rdf:RDF>
118:
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Feedback

If you suspect the parser is in error, please enter an explanation below and then press the **Submit problem report** button, to mail the report (and listing) to www-rdf-validator@w3.org